

LESSON 5-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Trapezoids by Decomposing

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles.

REMEMBER

- **Decompose** a trapezoid into a rectangle and one or two triangles.
- Find the area of each piece, then **add** the areas.
- When you cut off a rectangle, the triangle bases together equal the **difference** of the two parallel sides.

WORKED EXAMPLE

Parallel sides 6 m and 12 m, height 5 m. Find the area.

Step 1 Rectangle: $6 \times 5 = 30$

Step 2 Triangle bases: $(12 - 6) \div 2 = 3$ m each

Step 3 Triangles: $2 \times (\frac{1}{2} \times 3 \times 5) = 15$

Step 4 Add: $30 + 15 = 45$

Answer **45 m²**

WORD BANK

trapezoid – 4 sides, one pair of parallel sides

parallel – always the same distance apart

decompose – cut into smaller shapes

triangle area – $\frac{1}{2} \times \text{base} \times \text{height}$

PRACTICE

show your work • label units

1. The trapezoid is cut into a rectangle and two matching triangles.

Step 1 First find the rectangle: top $\times$ height.

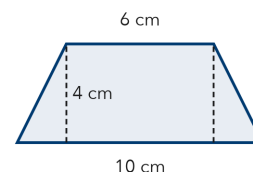
Step 2 Then find one triangle: $\frac{1}{2} \times \text{base} \times \text{height}$.

Step 3 Add the rectangle and both triangles.

a. Area of the rectangle _____ cm²

b. Area of one triangle _____ cm²

Area of trapezoid _____ cm²

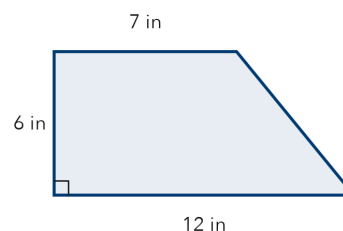


2. Decompose the trapezoid into a rectangle and one triangle. Find its area.

Step 1 Rectangle: short side $\times$ height.

Step 2 Triangle base: $12 - 7$.

Step 3 Add the two areas.



Show your work.

Answer _____ in²

3. **Skate Ramp.** A skate ramp side is a trapezoid with one **square corner**. The parallel sides are 3 ft and 8 ft. The height is 4 ft. Cut it into a rectangle and a triangle. Find the area.

Step 1 Rectangle: 3×4 .

Step 2 Triangle base: $8 - 3$.

Step 3 Add the two areas.

Show your work.

Answer _____ ft²

4. **Table Top.** A table top is a **trapezoid**. The parallel sides are 5 ft and 11 ft. The height is 3 ft. Cut it into a rectangle and two **matching** triangles. Find the area.

Step 1 Find the rectangle.

Step 2 Split $11 - 5$ into two equal triangle bases.

Step 3 Add all three pieces.

Answer _____ ft²

5. **Find the Error.** A trapezoid has parallel sides 9 cm and 15 cm and height 4 cm. Tomás says each of its 2 matching triangles has a base of 6 cm. Explain his mistake and find the area.

Frame Tomás is wrong because the two triangle bases together are ___, so each one is ___ cm.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #___

AT HOME Find a trapezoid shape at home or outside (a lampshade side, a bag, a roof). Show someone how to cut it into a rectangle and triangles.

LESSON 5-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**The Trapezoid Area Formula**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a trapezoid using the trapezoid area formula.

REMEMBER

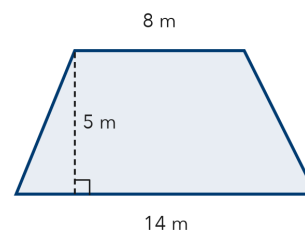
- **$A = \frac{1}{2} \times (b_1 + b_2) \times h$** . The bases b_1 and b_2 are the **parallel** sides.
- Add the bases first, then multiply by the height h , then take half.

WORKED EXAMPLE

Parallel sides 7 cm and 11 cm, height 6 cm. Find the area.

Step 1 Add the bases: $7 + 11 = 18$ **Step 2** Multiply by the height: $18 \times 6 = 108$ **Step 3** Take half: $108 \div 2 = 54$ **Answer** **54 cm²****WORD BANK****base** – one of the 2 parallel sides**height** – straight distance between the bases**formula** – $A = \frac{1}{2} \times (b_1 + b_2) \times h$ **sum** – the answer when you add**PRACTICE***show your work • label units*

- 1.**
- Find the area of the trapezoid.

Step 1 Add the bases: $8 + 14$.**Step 2** Multiply by the height.**Step 3** Take half.*Show your work.***Answer** _____ m²

- 2.**
- Find the area of each trapezoid with bases
- b_1
- and
- b_2
- and height
- h
- .

a. 4 ft, 10 ft; $h = 6$ ft _____ ft²**b.** 5 in, 9 in; $h = 8$ in _____ in²**c.** 6 m, 12 m; $h = 3$ m _____ m²**d.** 3 cm, 7 cm; $h = 9$ cm _____ cm²

- 3. Garden Plot.**
- A garden is a
- trapezoid**
- . The parallel sides are 16 ft and 24 ft. The height is 10 ft. One bag of mulch covers 25 ft
- ²
- . How many bags do you need?

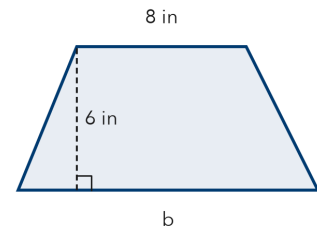
Step 1 Find the area with the formula.**Step 2** Then divide the area by 25.**Answer** _____ bags

4. **Missing Base.** The area of the trapezoid is 60 in^2 . Find the missing base b .

Step 1 Double the area: 2×60 .

Step 2 Divide by the height. This gives $b_1 + b_2$.

Step 3 Subtract the base you know.



Show your work.

Answer _____ in

TEST PRACTICE

choose the best answer

5. A trapezoid has parallel sides of 6 cm and 10 cm and a height of 5 cm. What is its area?

A 21 cm^2 **B** 40 cm^2 **C** 80 cm^2 **D** 300 cm^2

6. Which expression finds the area of a trapezoid with bases 3 m and 9 m and height 4 m?

Step 1 The bases are added first, inside ().

Step 2 Then multiply by the height and take half.

A $\frac{1}{2} \times 3 \times 9 \times 4$ **B** $(3 + 9) \times 4$ **C** $\frac{1}{2} \times (3 + 9) \times 4$ **D** $\frac{1}{2} \times 3 + 9 \times 4$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Measure a trapezoid shape at home, like the side of a bag. Estimate its area with the formula.